White Band Associates

WHITE BAND ASSOCIATES

Cyber Security

Summer Internship 2026

Networking Task 01

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Domain : Cyber Security
Week 1 - Networking

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INTRODUCTION

Computer networking is the process of connecting devices to enable communication and data exchange. It forms the foundation of modern technology, allowing systems to access the internet, share resources, and communicate efficiently. Understanding networking concepts such as IP addresses, MAC addresses, DNS, and network connectivity is essential for cybersecurity professionals. This task focuses on exploring basic network configuration and performing connectivity tests using networking tools and commands.

OBJECTIVE

The objective of this task is to understand basic networking concepts and gain practical experience in identifying network configuration details such as IP Address, MAC Address, Default Gateway, and DNS Server. This task also aims to verify network connectivity using Ping and Traceroute commands and develop a fundamental understanding of network communication.

SYSTEM INFORMATION

Parameter	Details
Name	Bhakti Parhad
Operating System	Windows 11
Virtualization Software	VMware Workstation
Linux Distribution	Kali Linux
Network Adapter	Intel(R) Wi-Fi 6 AX203
Task Domain	Networking
Internship Program	White Band Associates Summer Internship 2026

NETWORK CONFIGURATION DETAILS

The following network configuration details were obtained using the ipconfig /all command in Windows Command Prompt.

Parameter	Value
Hostname	LAPTOP-UPABTK5H
IPv4 Address	10.180.51.70
MAC Address (Physical Address)	xx-xx-xx-xx-xx-xx
Default Gateway	10.180.51.197
DNS Server	10.180.51.197

Description :

1. **Hostname :** Identifies the computer on the network.
2. **IPv4 Address :** The unique address assigned to the device for communication within the network.
3. **MAC Address :** A unique hardware address assigned to the network interface card.
4. **Default Gateway :** The router that forwards network traffic to external networks and the internet.
5. **DNS Server :** Translates domain names such as google.com into IP addresses.

Figure 1 : Hostname Information

```
C:\Users\bhakt>ipconfig /all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : LAPTOP-UPABTK5H
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . :
   IP Routing Enabled. . . . . . . . :
   WINS Proxy Enabled. . . . . . . . :
```

Figure 2: IPv4 Address and MAC Address

```
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : VMware Virtual Ethernet Adapter for VMnet1
   Physical Address. . . . . . . . . :
   DHCP Enabled. . . . . . . . . . . :
   Autoconfiguration Enabled . . . . :
   Link-local IPv6 Address . . . . . :
   IPv4 Address. . . . . . . . . . . : 192.168.27.1(Preferred)
   Subnet Mask . . . . . . . . . . . :
   Lease Obtained. . . . . . . . . . :
   Lease Expires . . . . . . . . . . :
   Default Gateway . . . . . . . . . :
   DHCP Server . . . . . . . . . . . :
   DHCPv6 IAID . . . . . . . . . . . :
```

Figure 3: Default Gateway and DNS Servers

```
ireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . :
   Physical Address. . . . . . . . . :
   DHCP Enabled. . . . . . . . . . . :
   Autoconfiguration Enabled . . . . :
   IPv6 Address. . . . . . . . . . . :
   Temporary IPv6 Address. . . . . . :
   Link-local IPv6 Address . . . . . :
   IPv4 Address. . . . . . . . . . . :
   Subnet Mask . . . . . . . . . . . :
   Lease Obtained. . . . . . . . . . :
   Lease Expires . . . . . . . . . . :
   Default Gateway . . . . . . . . . : fe80::147d:a7ff:fe22:b497%10
                                       10.180.51.197
   DHCP Server . . . . . . . . . . . :
   DHCPv6 IAID . . . . . . . . . . . :
   DHCPv6 Client DUID. . . . . . . . :
   DNS Servers . . . . . . . . . . . : 10.180.51.197
                                       2409:40c2:6420:17c2::de
   NetBIOS over Tcpip. . . . . . . . : Enabled
```

NETWORKING CONCEPTS

IP Address

An *IP (Internet Protocol)* Address is a unique numerical identifier assigned to a device on a network. It enables communication between devices and helps in sending and receiving data.

MAC Address

A *MAC (Media Access Control)* Address is a unique hardware address assigned to a *network interface card (NIC)*. It is used to identify devices within a local network.

Default Gateway

A Default Gateway is a network device, usually a router, that forwards traffic from a local network to external networks such as the Internet.

DNS Server

A *DNS (Domain Name System)* Server translates human-readable domain names such as google.com into IP addresses that computers can understand.

Public IP Address

A Public IP Address is assigned by an Internet Service Provider (ISP) and is used for communication over the Internet.

Private IP Address

A Private IP Address is used within a local network and cannot be accessed directly from the Internet. *Example : 10.180.51.70.*

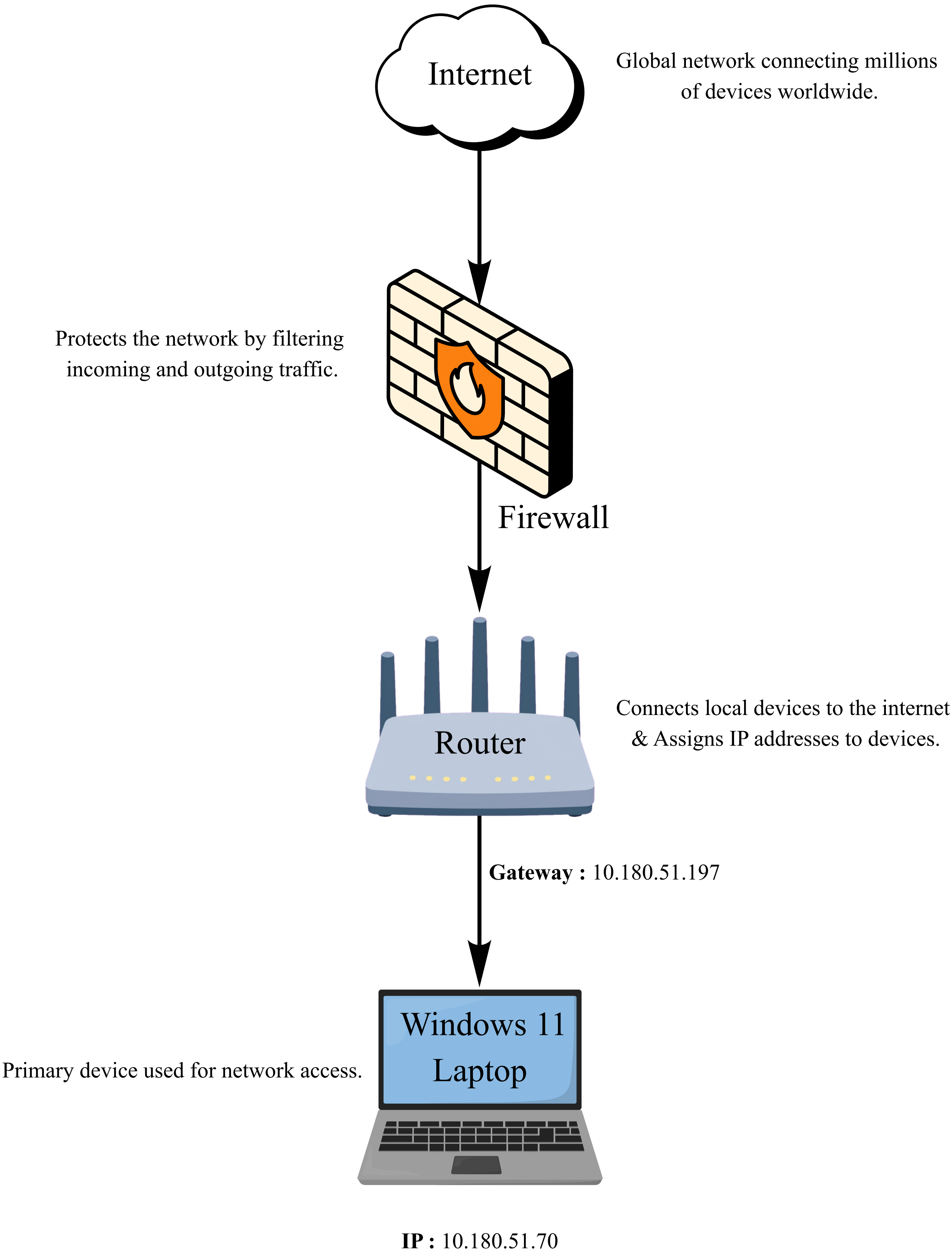
Ping

Ping is a network diagnostic tool used to test connectivity between two devices and measure response time.

Traceroute

Traceroute is a network diagnostic tool used to identify the path that data packets take from the source device to the destination.

NETWORK DIAGRAM



PING TEST

Command Used

ping google.com

Purpose

The Ping command is used to test connectivity between the local device and a remote host. It helps determine whether the destination is reachable and measures the response time.

Observation

The command successfully received replies from Google's server, indicating that the internet connection was active and functioning correctly. The response time was displayed in milliseconds (ms), confirming successful communication with the destination host.

Figure 4 : Ping Command Output

```
C:\Users\bhakt>ping google.com

Pinging google.com [2404:6800:4002:80b::200e] with 32 bytes of data:
Reply from 2404:6800:4002:80b::200e: time=70ms
Reply from 2404:6800:4002:80b::200e: time=85ms
Reply from 2404:6800:4002:80b::200e: time=61ms
Reply from 2404:6800:4002:80b::200e: time=69ms

Ping statistics for 2404:6800:4002:80b::200e:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 61ms, Maximum = 85ms, Average = 71ms
```

TRACEROUTE TEST

Command Used

tracert google.com

Purpose

The Traceroute command is used to identify the path taken by network packets from the source device to the destination server. It displays the intermediate routers (hops) through which the packets travel.

Observation

The command successfully displayed multiple hops between the local system and Google's server. Each hop represented a router or network device through which the packets passed before reaching the destination.

Figure 5 : Traceroute Command Output

```
C:\Users\bhakt>tracert google.com

Tracing route to google.com [2404:6800:4002:80b::200e]
over a maximum of 30 hops:

  1     3 ms     3 ms    40 ms  2409:40c2:6420:17c2::de
  2     *        *        *     Request timed out.
  3    70 ms    35 ms    38 ms  2405:200:5205:28:3925::ff21
  4     *        *        *     Request timed out.
  5     *        *        *     Request timed out.
  6    70 ms    30 ms    35 ms  2405:200:801:6000::16
  7     *        *        *     Request timed out.
  8     *        *        *     Request timed out.
  9    57 ms    40 ms    71 ms  2404:6800:8281:2c0::1
 10    78 ms    93 ms    43 ms  2404:6800:8281:2c0::1
 11    48 ms    54 ms    64 ms  2001:4860:0:1::269e
 12    61 ms    37 ms    43 ms  2001:4860:0:1::7140
 13    77 ms    34 ms    38 ms  2001:4860::c:4004:53ca
 14    75 ms    70 ms    55 ms  2001:4860::c:4004:153f
 15    76 ms    88 ms    91 ms  2001:4860:0:1::75b6
 16    98 ms    72 ms    76 ms  tzdela-bb-in-x0e.1e100.net [2404:6800:4002:80b::200e]

Trace complete.
```

LINUX (KALI) VERIFICATION

Objective

To verify network configuration and network connectivity using networking commands in Kali Linux.

Commands Used

- ip addr
- ping google.com
- traceroute google.com

Purpose

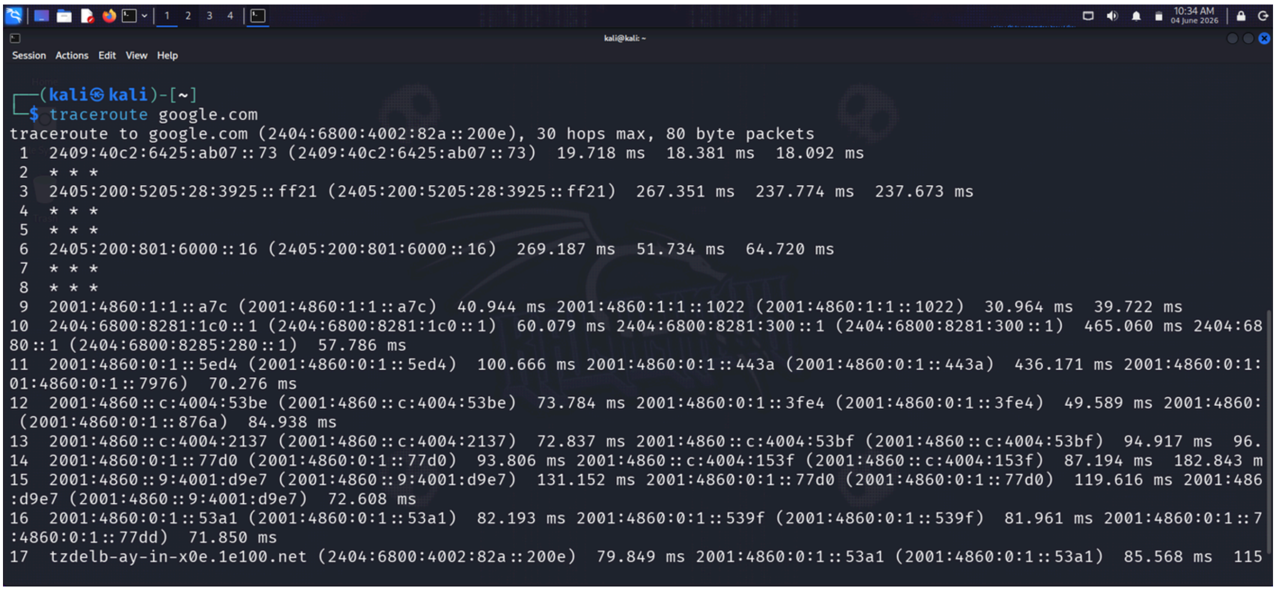
- *ip addr* was used to view network interface details, IP addresses, and MAC addresses.
- *ping* was used to verify connectivity between the Kali Linux system and Google's server.
- *traceroute* was used to identify the path taken by packets from the Kali Linux system to the destination server.

Observation

The Kali Linux virtual machine successfully displayed its network configuration, including the assigned IP address and network interface information. The ping test received successful replies from Google's server, confirming active internet connectivity. The traceroute command displayed the network path and intermediate hops between the source and destination.

Figure 6 : Kali Linux Network Configuration and Ping Test

Figure 7 : Kali Linux Traceroute Verification



LEARNING OUTCOMES

Through this task, I gained practical knowledge of basic networking concepts and commands. The key learning outcomes are :

- Learned how to identify network configuration details using the ipconfig /all command.
- Understood the purpose and importance of IP Addresses, MAC Addresses, DNS Servers, and Default Gateways.
- Learned the difference between Public and Private IP Addresses.
- Performed network connectivity testing using the ping command.
- Analyzed the path taken by network packets using the tracert command.
- Created a basic network diagram to understand network communication flow.
- Gained hands-on experience with networking tools in both Windows and Kali Linux environments.
- Improved documentation and reporting skills by organizing findings and screenshots in a structured format.

CONCLUSION

This task provided practical exposure to fundamental networking concepts and tools. By examining network configuration details, performing connectivity tests using Ping and Traceroute, and creating a network diagram, I gained a better understanding of how devices communicate within a network and access the Internet. The task also helped strengthen my knowledge of IP addressing, DNS, MAC addresses, and network troubleshooting, which are essential foundations for Cyber Security and IT.